SESAR SECURITY 2020: HOW TO EMBED AND ASSURE SECURITY IN SYSTEM-OF-SYSTEMS ENGINEERING?

Rainer Koelle and Martin Hawley, EUROCONTROL, Brussels, Belgium

Abstract

Significant efforts are underway to modernise global air traffic management systems. This will result in a level of connectivity between different systems that has never before been achieved, creating a 'system of systems' that requires a very high level of dependability. To achieve this dependability, security must be designed-in rather than relying on hardening of systems post implementation. This is also expected to minimise the costs of security. The European ATM modernisation programme, SESAR, has been taking a systems engineering approach to ensure that R&D is rapidly transferred to industrialisation and then deployment. This approach has been applied to security, organised around 'operational focus areas' that represent discrete operational improvements. In such a complex programme, traditional risk analysis and mitigation was thought to be limited when attempting a system wide coherence for security. Hence the SESAR programme is adopting a 'security case' approach, drawing on lessons learned in safety cases, validation & verification of ATM R&D, and aligned to the system engineering approach.

Introduction

Transformation of Air Traffic Management

With the beginning of the 21st century, aviation is undergoing a fundamental transformation driven by a desire to increase its performance across the board. Within the domain of air traffic management, (ATM) this has led to comprehensive programmes to modernise the ATM 'System', principally SESAR, NextGen and CARATS. Under these programmes, new operational concepts and technical enablers are being developed to transform the industry, leading to the vision of a sustainable and performance based ATM system. The European programme, SESAR, is known as the technological 'pillar' of the 'Single European Sky' (SES). The SES has been reforming European air traffic management since 2004 and has the ambition to halve unit costs, treble capacity, increase safety by a factor of 10 and decrease by 10% the environmental impact of flights. These performance targets and others are set out in the European ATM Master Plan [1]. Security is an important element of the Master Plan, which sets out a roadmap for security, as both an enabler for aforementioned performance targets and as a key performance area in itself.

Distributed and Highly Interconnected System of Systems

Air traffic management infrastructure has become increasingly connected, relying on private networks to coordinate between air traffic control centres and share resources such as radar feeds. Under the modernisation plans, this connectivity will greatly increase to support new concepts, and ultimately 4D trajectory management. Hence the future modernised air transportation/management system will become a distributed and highly interconnected 'system of systems'. This poses both challenges and opportunities for security management:

- challenges from the need to assure the security of such mission-critical or high-reliability systems;
- opportunities from the prospect of 'designing-in' security on a system-wide scale.

Like any other design requirement, it is well understood that security needs to be designed into a system rather than being bolted on as an afterthought [2]. This serves to minimise implementation costs and create a high degree of resilience throughout the system life-cycle.

The Challenges of Designed-In Security

The security by design approach in SESAR creates the challenge to embed security into the programme's constituent projects, combining the disciplines of security risk assessment with the more traditional concept and requirements development

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process. Despite a greatly altered security landscape in recent years, the consideration of security aspects throughout the development process is too often given low priority, or is simply misunderstood. This is beginning to change, with higher recognition of the importance of establishing and assuring resilience and security of mission-critical systems: including avionics and ground-based ATM/CNS systems. However, modelling security assurance cases of system of systems adds another layer of complexity. While a system may be deemed secure by itself in a particular configuration, the introduction of other systems into that environment increases the complexity of the assurance model and must be considered and evaluated as part of a larger system.

This paper is structured as follows: first we provide an overview of SESAR, Aviation Security and Air Navigation Security, and the high-level security management approach in SESAR. Next, the principal interplay between SESAR Engineering processes and security tools will be discussed. We then present the current status and initial findings, before we close this paper and identify upcoming work.

Background

SESAR

SESAR - the Single European Sky ATM Research Programme - is the means of defining, designing and delivering the operational and technological changes necessary to achieve a more efficient, better integrated, more economical, safer and more environmentally sustainable European ATM infrastructure by the year 2020. The Programme is managed by the SESAR Joint Undertaking (SESAR JU) and comprises three distinct phases: Definition, Development and Deployment. The definition phase, completed in 2008, delivered the European ATM Master Plan [2]. This Master Plan is currently being refined as part of the on-going developments. SESAR is now in the development phase, aiming to develop the concepts and technologies and other associated requirements defined in the Master Plan. Following this, a future Deployment phase (2016-2020) will see the large scale production and implementation of the new air traffic management infrastructure.

The European ATM Master Plan identified a sequence of improving service levels (steps 0-5) and structured these into 3 implementation packages (IP 1-3). These packages relate to three major operational steps: 1) time-based operations; 2) trajectory-based operations; and 3) performance-based operations. Further definition within each step is provided through detailed 'Operational Improvement' steps and their associated enablers. There have also been concerted efforts to link the envisaged SESAR deployments to ICAO's concept of 'Aviation System Block Upgrades' [3]. This draws on the implementation plans of the ICAO regions and ATM modernisation programmes to create a sequence of 4 blocks; each describing the operational improvement, required equipment/systems, standards and procedures, and each to be supported by a business case. Hence, in the coming years, air traffic management will evolve as a series of steps corresponding to improvements in air navigation service levels based on associated enablers, including aircraft systems.

Because of its large scale, SESAR incorporates a number of programme management disciplines within an overarching framework. This aims to ensure coherence between the ATM Target Concept, Business Case, System Architecture, and the operational and technically validated project deliverables. To make sense of the overall vision and components of SESAR, the 'system of systems' is being defined in a single European ATM Enterprise Architecture. The European ATM Enterprise Architectue comprises 5 major views: 1) business view; 2) programme view; 3) organisational view; 4) system view; and 5) technical view.

Because SESAR deliverables will ultimately support operational services, there is a need to ensure coherence between the operational and technical deliverables.

This has been achieved by linking the programme management hierarchy, defined by Work-Packages, Sub Work-Packages and Projects, to an operational, deployment focused, hierarchy of Operational Packages/Sub-packages (c.f. Figure 1) and 'Operational Focus Areas'. Operational Focus Areas (OFAs) comprise a limited set of projects concerned with dependent operational and technical improvements [4]. OFAs link interrelated Operational Improvements (OIs) and enablers to

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address specific performance requirements. This structure is also intended to result in coherent and integrated validation and verification activities and associated results.

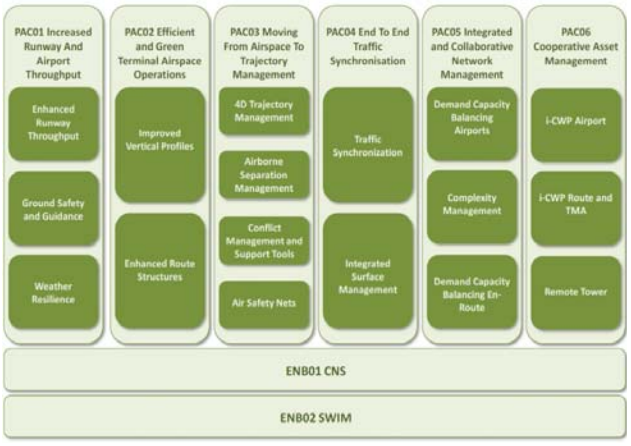


Figure 1. SESAR – Operational Focus Areas

Aviation Security – Air Navigation Security

The security of aviation has been a growing concern since the first hijackings took place in the early 20th century. Undoubtedly September 11th 2001 changed the aviation world forever, and how we approach the security of aviation. In the aftermath of 9/11 increased efforts in the identification and security of critical infrastructures were undertaken by all States. Since 9/11, States have been reappraising what they consider as critical infrastructure, and for many, air transportation is included, with Air Navigation Services (ANS) inheriting this criticality. To add to this view, there is growing recognition of the enabler role of ANS, and its impact on the overall performance of the transport network. A prominent example is the recent ash cloud 'crisis' where aviation was significantly hampered. Though this had an environmental cause, the order of magnitude and associated ripple effects serves as a reminder for the potential consequences of a large-scale impact; such as could be caused by an attack to networked infrastructure under the future modernised ATM system.

The International Civil Aviation Organization defines "Aviation Security" in Annex 17 as "Safeguarding civil aviation against acts of unlawful interference. This objective is achieved by a combination of measures and human and material resources." [5] These measures can be correlated to different aspects of the aviation supply chain; so that to secure the safe, orderly and efficient flow of air traffic, the interplay of people, technology, and processes needs to address security [6].

With the latest update of Annex 17, 2011, ICAO recognises the on-going transformation of aviation and future challenges of aviation security [5]. For example, Air Traffic Service providers are now specifically addressed in Annex 17 and an initial recommendation for cyber security has been included. Nevertheless, the security of Air Navigation Services (ANS) is a developing topic, particularly when concerning future modernisation plans. Looking to the future, the ICAO vision for Global Air Traffic Management is for "...an interoperable global air traffic management system, for all users during all phases of flight, that meets agreed levels of safety, provides for optimum economic operations, is environmentally sustainable and meets national security requirements." [7].

The expectation for ATM security is authoritatively defined in ICAO Document 9854 [7], which lists security among 11 global (high level) performance expectations: "Security refers to the protection against threats... Adequate security is a major expectation of the ATM community and of citizens. The ATM system should therefore contribute to security, and the ATM system, as well as ATM related information, should be protected against security threats...In the event of threats to aircraft or threats using aircraft, ATM shall provide responsible authorities with appropriate assistance and information.

In spite of ICAO Doc 9854, at the time being there is no globally accepted definition for Air Navigation (or ATM) Security. Efforts are on-going at ICAO level to develop an 'ATM Security Manual', based on the experience of EUROCONTROL and the FAA in this field. The current definition for ATM Security - an input to the aforementioned work - is based on a EUROCONTROL workshop on ATM Security held in December 2006. The workshop developed a common understanding on ATM Security being concerned with "… threats that are aimed at the ATM System directly, such as attacks on ATM assets, or where ATM plays a key role in the prevention or response to threats aimed at other parts of the aviation system (or national and international

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assets of high value) and limiting their effects on the overall ATM Network."[8,9]

This understanding has been the basis for the SESAR Definition Phase and is now further refined as part of the Development Phase. Linking Air Navigation Security with outcomes, the following is referred to as the dual nature of air navigation security:

- Resilience / self-protection. The (primary) scope of ATM Security can be characterized by ensuring the resilience, integrity and availability of the ATM and CNS assets and air navigation services to enable the safe, secure, economic, expeditious and orderly flow of traffic throughout all phases of flight.
- Collaborative support: The (secondary) scope of ATM Security aims at the provision of (support) services to other authorities / organisations dealing with aviation security.

The EUROCONTROL Guidance to the European ATM Master Plan [10] specifies the primary outcome of ATM Security as ensuring the continued provision of air navigation services. Disruptions shall be restricted to 'significant events' and limited to a subset of system functions and capabilities. Degradation of service levels shall be recovered within a proportionate period of time.

It is important to further define what is meant by 'Collaborative Support'. A fundamental role of any Government is the security of its citizens. As a result, national defence requirements determine the role and mission of the military forces, which include Airspace Security. NATO has defined Airspace Security as the "safeguarding of the airspace or area of responsibility from unauthorized use, intrusion, illegal activities or any violation". Collaborative Support as introduced above is that part of ATM security directed to support national authorities or entities in their airspace security or homeland security mission. These authorities include not only military command, but also national / regional crisis management authorities, police and law enforcement, local emergency forces (in the case of ground interventions), etc.

A growing number of States are implementing an outcome or performance based regulatory framework. With a view to the transportation sector in general or air transportation in particular, several States (e.g. EU, Canada, New Zealand) have already implemented legislation and regulation mandating Security Management Systems (SecMS). Security Management Systems are viewed as the framework within which other security activities and standards are incorporated, for example security risk assessment, the identification of security measures, the development of security programmes or the compliance with regulation and standards. In the European context, the Common Requirements for the security of air navigation have been specified in EC Regulation 1035/2011 repealing the former EC Regulation 2096/2005 [11,12]. The Common Requirements require air navigation service providers (ANSPs) to establish a SecMS and ensure the security of their staff, facilities and operational data. Adherence to the Common Requirements is a requirement to be a 'certified' air traffic service provider under the 'Service Provision' Regulation, EC 550/2004 [13]. The existing regulatory framework is a driver for SESAR.

Security Management in SESAR

The programme structure of SESAR includes what are known as 'transversal areas', which cut across and support the main operational and technical work packages. The transversal areas (TAs) include Safety, Security, Environment, Human Performance and the Business Case; all grouped under Work Package 16. Table 1 shows how security is structured in SESAR under 5 projects (16.02.0x) plus a coordinating project (16.06.02). These projects collectively cover the development of methods and tools and tailoring of the body of knowledge to the needs of SESAR. The support and coordination function aims to ensure that security management and assurance activities are correctly applied and that a consistent approach is taken throughout the work programme. The level of support provided ranges from passive (e.g. awareness and guidance material) to active (e.g. coaching).

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Table 1. Security Thread – Project Scope

Project	Scope / Focus
16.02.01	Security Management system / framework spanning many organisations and life-cycle phases (focus design and development)
16.02.02	Holistic evaluation scenarios to support structured assurance case building and project validation activities
16.02.03	Security risk management method alignment with SESAR system engineering
16.02.04	Security validation and verification techniques
16.02.05	Identification and gap analysis of existing regulatory framework, standards and best practices, identification and development of baseline measures for SESAR and proposal for ICAO/EC changes
16.06.02	Support and Coordination Function providing passive and active awareness and coaching to SESAR programme

The SESAR programme does not itself have a security management system (SecMS), although it is recognised that the deployed outputs of SESAR will be implemented under security management systems as required by National Supervisory Authorities. Neither does SESAR impose a single SecMS for implementation by all SESAR Members or stakeholders.

There are, however, features of the SESAR security methodology that accord with the Security Management System approach. In particular, the SESAR methods adopt the principles of the Deming cycle, starting with policy:

- Policy – endorsed and enforced by the SESAR JU, with roles and responsibilities assigned to OFA Managers and WP 16.06.02.
- Plan – which is done as part of the system engineering approach with operational and technical projects, with WP 16.06.02 supporting prioritisation and cross-checking of activities.
- Do – programme / OFA applies security reference material undertakes security risk assessment and assurance activities.
- Check – primarily a role of WP 16.06.02 role on behalf of the SJU.
- Act – if required remedial action will be identified as part of the formal SE reviews or (informally) during development (i.e. coaching & hands-on support).

'Modelling' Approach

Security and Dependability/Security System Engineering

In developing the future ATM system of systems, the scope and complexity of SESAR demands a system engineering approach, so that requirements are properly captured and are traceable, and dependencies between system elements are identified and addressed. Systems engineering is a methodical, disciplined approach for the design, realization, technical management, operations, and retirement of a system [14]. The ultimate goal of systems engineering is to design, develop and deploy systems that are dependable.

An important part of the systems engineering approach is to conduct phase reviews along the development lifecycle, to ensure that requirements are being defined and are feasible. In SESAR, the techniques for validating requirements are devolved from the European Operational Concept Validation Methodology, or E-OCVM [15]. This was specifically developed to support the efficient transfer of concepts from R&D through to operation and its principles have fed into the SESAR System Engineering approach and Validation and Verification strategy. E-OCVM defines a system's validation life cycle in eight phases (V0-V7), of which phases V1-V3 see the validation of concept and requirements. In SESAR terms, completion of validation phase V3 means that the concept is valid to proceed to the industrialisation phase and marks the completion of SESAR development on it. The System Engineering phase reviews tie in with the V1-V3 phases, acting as a gate decision to proceed to the next phase, whilst allowing some flexibility for validation activity to be carried over. This process is shown in Figure 2.

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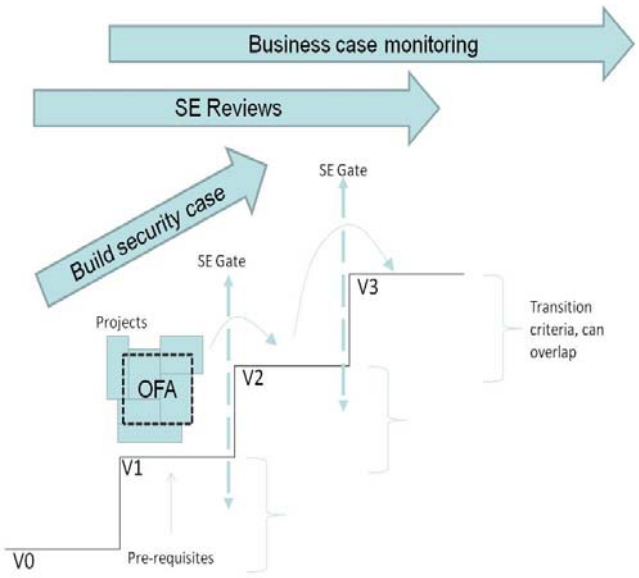


Figure 2. Interplay of SE Reviews and Cases

We introduced the notion of critical infrastructure and critical systems above. Criticality is an important consideration in risk assessment, and can be determined from the estimated impact of an attack on the system's primary/supporting assets. Criticality can be related to the business and operational architecture viewpoints:

1.) Business view: partial/complete outage or loss of air navigation services or functions results in loss of confidence in air transport, airspace user operations and ultimately in high economic losses.

2) Operational view: the objective of air navigation is the safe, orderly and efficient flow of air traffic; security impacts resulting in safety consequences (e.g. malfunctioning of safety nets) can result in loss of life, injury or damage to the environment; other mission-critical aspects can include significant impacts on capacity; untrustworthy aeronautical data can lead to significant increase of workload.

It is noteworthy to mention that collaborative support requires capabilities ranging from voice communication to system interoperability which themselves can be subject to failures; whereby mission objectives/goals would not be met.

The outcomes of the systems engineering approach should be a system that is, in essence, more dependable. Dependability has been defined by Avizienis et al [16], from which a range of system properties can be related to security. I.e., it is clear that security failures will have an impact on the following attributes: availability; readiness for correct service; reliability; continuity of correct service.; safety; absence of catastrophic consequences on the user(s) and the environment; integrity; absence of improper system alterations; maintainability; ability to undergo modifications and repairs. It must be noted that the classical interpretation of Security in aviation and ATM subsumes availability, reliability and integrity, which Avizienis refers to as a composite [16].

SESAR Security Policy, High-Level Objectives, SESAR Systems Engineering Approach and Deliverables

Security policy is an expression of the overall intent of an organisation with respect to its security, and may define objectives, commitment and specific responsibilities. The ultimate security goal of SESAR is to meet the envisaged performance in terms of self-protection/ resilience, and collaborative support. The SESAR security policy, currently under review, permits its policy statements to be flowed down to lower levels in the programme. This enables specific interpretation of policy to the different levels: work-packages, operational focus areas and individual projects. I.e., the top-level goal needs to be instantiated in and tailored to the high-level objectives for each Operational Focus Area. An example could be that OFA XYZ "shall ensure integrity of displayed and processed data."

The ultimate security goal of SESAR is to meet the envisaged performance in terms of self-protection / resilience and collaborative support. The following Table 2 provides an example of high-level security objectives derived for a specific OFA.

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Table 2. High-Level Security Objectives

ANS Security Focus Area	High-Level Security Objectives
Self-protection Resilience	• to ensure continued provision of Air Navigation Services and the efficiency of the mission / business trajectory during the partial loss of function A • to meet the regulatory requirements for quality or aeronautical data
Collaborative Support	• to harmonise and ensure pan-European provision of aviation security incident management support; • to ensure interoperability and information sharing with incident management stake-holders (e.g. military, law enforcement/police, aviation security crisis centres)

Structured Assurance Case For SESAR Security

Given the complexities discussed in this paper thus far, it was not believed that security could be designed-in on the basis of risk assessments alone. Other factors would need to be addressed to ensure that security was met at system and system of systems level, and progressively over the operational deployment timeframe. It is also desirable to minimise repetition of security assessments where assets are common to two or more operational focus areas. Hence it was decided to take a 'security case' approach, drawing on lessons learned from ATM safety cases but tailored to security. The case concept is derived from the legal practice of presenting a legal case in a courtroom. The principal approach is to present arguments showing that certain claims are supported by objective evidence. Reference [17] foresees the case concept to become widely applied, dealing with safety, security, business and environment cases.

To frame the security case, the 16.06.02 project team has determined a security 'argument' that the operation in question is resilient, such that the security risks have been reduced to an acceptable level. The argument is structured into seven sub-claims that are believed to be key in supporting the argument, as illustrated in Figure 3. Whilst it is planned that the case should be made at the level of Operational Packages, evidence for the case will be gathered at the level of Operational Focus Area (OFA).

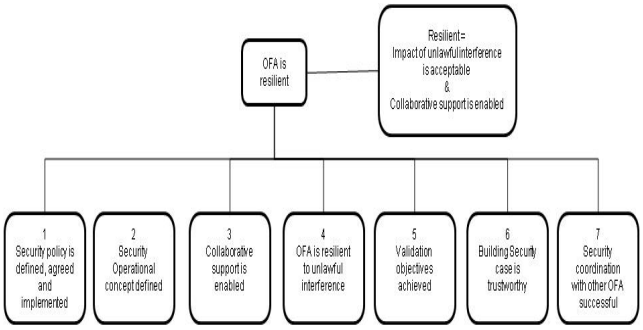


Figure 3. Security Case Pattern

Reading from left to right, the security case should firstly provide assurance that Security Policy is coherent with high level policy and appropriately cascaded to the OFA level. The next category of assurance is that security has been integrated into the operational concept, or that there is a 'security concept' for the OFA. This will assure that the operational concept will be resilient in the face of 'non-nominal' scenarios and that there has been due consideration of how the concept will contribute to collaborative support. It is crucial that these first two areas are covered early on in the development lifecycle for security to be truly designed-in. The next two sub-claims, Collaborative Support and Resilience, address the envisaged function of security in the intended environment. For collaborative support, it will be argued that the proposed arrangements are necessary and sufficient to support the operational concept. For resilience, the focus is on risk assessment and mitigation, and the claim is that the residual risk is acceptable. The Validation sub-claim is that the preceding claims have been validated, such as through analyses and experimentation that reflect the intended operating and threat environment. Next, the claim is made that the work done is trustworthy, having been carried out by experts and following the appropriate methods, standards and practices. The final sub-claim is that there has been coordination with other operational focus areas, to ensure no gaps or contradictory effects.

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Whilst the security of the system is ultimately the responsibility of those who operate it, and ANSPs in particular, the security case will also be subject to regulatory oversight. This oversight still requires definition by regulators. However, assuming the analogy with safety cases, the ultimate requirement will be to provide evidence to the sponsor (European Commission) and regulatory bodies (e.g. EASA, State National Supervisory Authorities) that SESAR outputs will be resilient and capable of providing the required collaborative support.

Having set out the security case approach, it is useful to outline some of the practical issues in embedding the approach throughout the SESAR programme. The overall programme security case is intended to be organised along the operational lines of the programme: Operational Work Package to Operational Focus Area. The SESAR ATM Security Case can therefore be seen as a compendium of 'package cases' per concept step. A further complication is that SESAR, with its strong delivery focus, creates programme outputs in a series of operational steps and releases. There are in fact two major development streams within the SESAR Programme: Firstly, 'Releases' are a sub-set of the SESAR Programme. A release comprises groups of projects delivering R&D results that will support a decision to move related activities to the industrialisation stage within a specific year. Secondly, there are projects that follow the E-OCVM phases and contribute to the later releases [18]. The consequence is that concepts may be at different stages of validation (V1-V3) for different steps in the concept. The security case is therefore framed with two dimensions, as in Figure 4:

- a claim / sub-claim dimension (vertical pattern), and
- a validation dimension (horizontal pattern).

The security work at the project level involves gathering the appropriate evidence for the claims, which may also be further broken down into sub-claims.

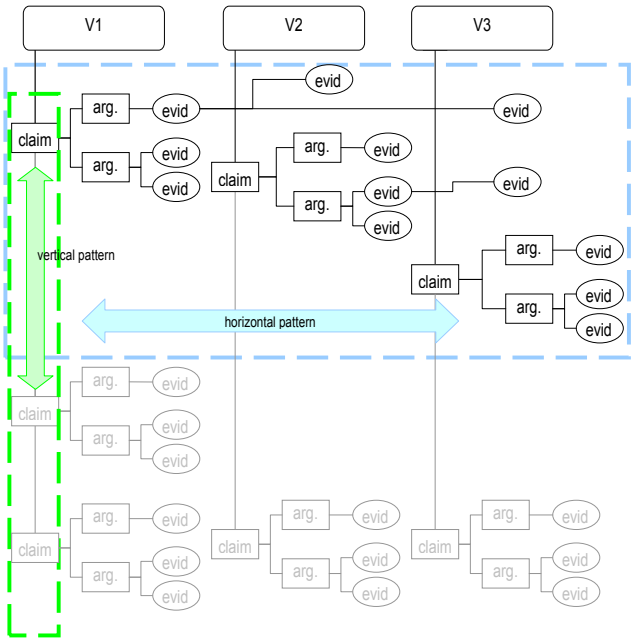


Figure 4. Vertical / Horizontal Case Patterns

Current Status and Initial Findings

SESAR Programme Appraisal

At the early stages of the programme there is a risk that the first SESAR releases will have immature security requirements, potentially adding to the costs of industrialisation. This risk is currently being addressed, where a particular mitigation is to prioritise the security work across the different operational focus areas. Prioritising OFAs from a security perspective has been a difficult task, because at first sight most seemed to have the potential to significantly disrupt operations in the event of a security incident. The security team therefore adopted a two-pronged approach:

- ranking OFAs according to impacts on safety, capacity and other key performance areas; and
- prioritising the set of OFAs that would cover all of the supporting assets.

The latter point is intended to ensure that security case evidence, and particularly for the resilience claim, is gathered for all supporting assets. I.e., as a minimum, the security case for the prioritised set of OFAs will include risk assessment and mitigation for all supporting assets.

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Table 3 presents a snapshot of the analysis matching OFAs to enablers / supporting assets. At this stage the many distinct enablers defined in SESAR have been loosely grouped, as the purpose is one of prioritisation. Further analysis of the relationships between OFAs and enablers is planned to understand the system-wide sensitivity of OFAs to enablers.

Table 3. Mapping Enablers - OFA

Enablers	Operational Focus Area			
	Trajectory Management Framework	SWIM	Dynamic Vortex Separation	...
Datalink	X	X	X	
FMS	X		X	
TWR Tools	X		X	
FDP mods	X	X		
ADS-B				

Compliance-Base Approach / Minimum Security Controls

Focusing limited resources on priorities is essential. Therefore a 'compliance based' approach has been introduced in SESAR. The intent is that not all SESAR deliverables / projects should undertake a full security-risk assessment and associated treatment activities. These projects will have to implement a certain suite of security best practices and associated safeguards, known as a 'minimum set of security controls'. This creates a potential conundrum, because it requires assumptions to be made on controls without risk analyses, along with judgements about the residual risk and whether risk appetite is likely to be met. Without a full risk assessment, a concern is that not all risks will be identified and treated, resulting in an incomplete set of requirements. However, it is believed that this concern is addressed by the integration of security into the validation life-cycle and system engineering reviews.

Claim-Argument-Evidence Patterns

The security case patterns have been discussed above. Table 4 provides a snapshot for the supporting claims. In this example, the argument that the security policy is 'defined, agreed and implemented' constitutes the subclaims. This paper revolves around the concept of embedding security practices into the system engineering. Accordingly, evidence for the security case are found throughout the system development life-cycle. In Table 4, evidence for different V-phases of the E-OCVM are listed.

Table 4. Security Case – Evidence Example

Sub-Claim	Evidence	
Security policy defined	V1	Roles and responsibilities
	V1	Covers people, procedures and equipment
	V1	Consistent with SESAR policy / interpretation on OFA level
	V1	Meets regulatory requirements
Security Policy agreed	V2	Signed off by OFA manager
	V3	Agreed with SJU
	V2	Distributed to all OFA Members
Security Policy implemented	V2	Policy Validation Objectives defined

Conclusions and Future Work

Conclusions

The current state of play is that SESAR completed its first Release in 2011 comprising a total of 18 OFAs. With the 4th quarter of 2011, all security R&D projects have entered their execution phase and initial versions of methods, tools and guidance (as reported throughout this paper) have been delivered. The current focus is on a trial-run of the assurance case methodology, initial security risk assessments and associated analyses.

The role of the transversal areas has been recognised on the programme level. The transversal areas are co-chairing the final and third Systems Engineering Review, the formal review to make a decision on the OFA to move into the next phase – pre-industrialisation. With this involvement, security considerations can be brought forward as part of the security case building. This includes the provision of guidance and associated reference material to enable the OFAs to integrate security risk assessment and assurance activities in their engineering and validation plans and present the required evidence

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during the reviews. The level of implementation in Release 1 varies strongly. This is partly linked with the infancy stage of SESAR and the successive enforcement of security deliverables and reference material for application by the SESAR programme.

Future work is targeted at validating novel 16.02.0x deliverables and integrate them into the Security Reference Material. This material comprises the suite of guidance for all projects and OFAs concerning security. The level of application will be closely monitored through the Release 2 Engineering Reviews and the 16.06.02 Support and Coordination Function. Release 2 poses challenges on resources as the number of OFAs roughly doubles.

A vital aspect of the future work will be the coordination with on-going standardisation activities in the field of aviation / ATM Security (e.g. EUROCAE WG72/RTCA SC-216). Another coordination thread is the acceptance and buy-in from regulatory bodies (e.g. EASA and NSAs). Synchronisation and compliance with ICAO work (ATM Security Manual, and Cyber Security). With a view to the latter, initial coordination with NextGen is initiated. These topics require globally accepted measures, practices and standards.

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Acknowledgements

The authors would like to thank Patrizia Montefusco and Edouard Painchault for the valuable comments during the drafting phase of this paper. Chris Machin was pivotal in the development of the structured assurance case method. Furthermore, the authors would like to thank all 16.06.02 Members for the discussions concerning the development of the SESAR Security Reference Material. These contributions are reflected throughout this paper.

Email Address

The corresponding author is: rainer.koelle@eurocontrol.int.

2012 Integrated Communications Navigation and Surveillance (ICNS) Conference

April 24-26, 2012

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